TEAM 25 · DECO3500 SOCIAL & MOBILE COMPUTING

NutriCart Hub

One shared grocery system for student share houses, from the kitchen to the supermarket aisle.

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Process, research & code

THE PROBLEM

- ✕ Requests get buried in the group chat
- ✕ Things get bought twice
- ✕ Splitting costs is awkward
- ✕ Food apps focus on one person and tell them what to eat

OUR OPPORTUNITY

How might a household plan and shop together, without extra effort or judgement?

WHAT WE DID

7 story-based interviews

13 prototype sessions (app probe + cardboard Hub)

2 group tests with housemates

plus in-store shadowing, a literature review of 8 papers and studio critique, across 3 prototype versions

THE EXPERIENCE · ONE HOUSEHOLD, TWO PLACES

Housemates plan at the Home Dock in the kitchen. Whoever shops takes the Cart Unit, so the house's list, requests and budget travel to the aisle.

🏠 Shared kitchen — HOME DOCK

Tablet in a kitchen stand · housemate tokens

- **G1 Plan together** — Tap your token, vote on meals, add what's missing — SAME TIME · SAME PLACE
- **G2 Leave a request** — "The cheap Greek yoghurt please", seen by whoever shops — DIFFERENT TIME · SAME PLACE
- **G6 Re-dock & settle** — What was bought, the budget, and who owes what — DIFFERENT TIME · SAME PLACE

→ **G3 Hand over the baton** "Jay is shopping now"

G4 Ask home — One quick question, answered with a tap — SAME TIME · DIFFERENT PLACES

re-dock ←

🛒 Supermarket aisle — CART UNIT

Phone on a basket or trolley mount

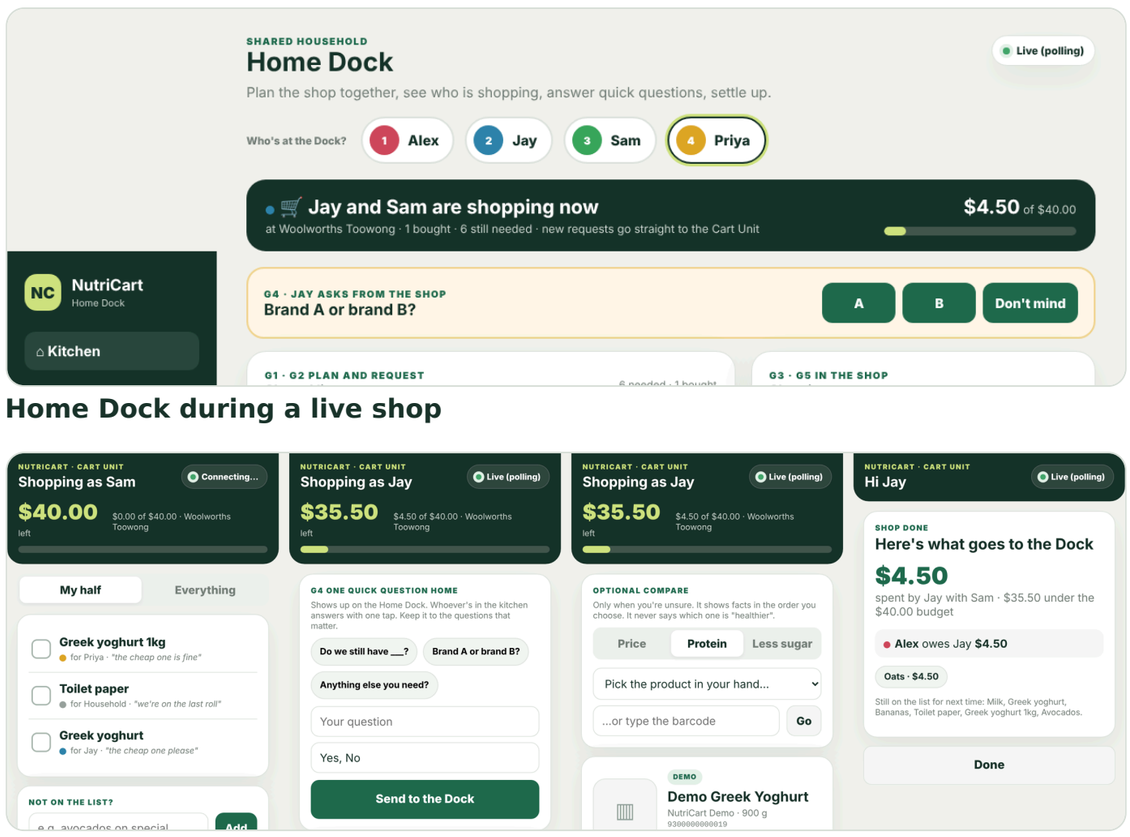
- **The household's list** — Requests, notes and "already at home?" flags travel with you — CONTEXT: IN STORE
- **Running budget** — Money left is the biggest thing on screen — CONTEXT: IN STORE
- **G5 Split Shop** — Two phones share one list and tick off live — SAME TIME · SAME PLACE

Where each moment sits on the time-space matrix

	Same time	Different time
Same place	G1 Plan together · G5 Split Shop	G2 Request · G3 Hand over · G6 Re-dock
Different place	G4 Ask home	G3 Shopping-now status, seen from home

Social: supports coordination and awareness the house already does, instead of a feed or leaderboard.
Mobile: location-dependent. Planning mode at home, a minimal shopping mode in the aisle.

Not: a social feed, a leaderboard, or a health score.



Home Dock during a live shop

Cart Unit: list, budget, compare

FROM RESEARCH TO DESIGN: WHAT WE HEARD → WHAT WE BUILT

INTERVIEWS

"99 out of 100 times" it's the price. R1-P02

The budget is the biggest number in store

→

V1 APP PROBE

A group goal felt like *"essentially telling me that I'm fat."* AP-P09

No scores, calories or leaderboards

→

V2 CARDBOARD HUB

Constant alerts would *"feel like noise, not help."* HP-P02

No push alerts; one deliberate "Ask home"

→

WEEK 9 STAND-UP

Staff: *"the social aspect is weak."*

Six moments where housemates act together

→

V3 LIVE PROTOTYPE

"I don't have to remember what my roommate texted me." HP-P02

One shared list, live across devices

WHAT TESTING SHOWED

2 group sessions with 4 students who live together · Week 12

4/4 named the budget, Compare or deals as the best part

The group chat is our real competitor

"Feels like another group chat." G01-P1 Another house valued seeing what's at home, so nothing is bought twice.

The social value is the live moment

"The only time I can see myself using this is when I'm talking to someone in the grocery store." G01-P2

First use is hard

All four needed help to navigate at first.

Next: lead with the live in-store moment · work alongside the group chat · simpler first use

DESIGNED WITH CARE

- ✓ Facts, not verdicts
- ✓ Shows *that* you're shopping, not where
- ✓ Aliases only, no accounts
- ✓ Household data stays local and is wiped on reset

Time-space matrix · **3C model** · **Location framework** · **Olsson et al. 2020** · **Schmidt 2002** · **Grudin 1994** · **Houde & Hill 1997**

TRY IT HERE: plan at the Dock → shop with the Cart Unit → watch the budget

Ethics, process & code: scan the QR